



Time: 5 minutes

Outcome: Analyze Short-Run Production & Diminishing Marginal Product.

Difficulty: Intermediate

Short-Run Production & Diminishing Marginal Product

THE CORE IDEA

In the short run, at least one input is fixed. Marginal product is the extra output from one more unit of a variable input. With fixed equipment, additional workers may eventually add less output than earlier workers. Diminishing marginal product means the addition falls; total product can still rise.

HOW TO RECOGNIZE IT

- Identify the fixed input and the variable input.
- Marginal product is the change in total product from one more worker.
- Diminishing marginal product begins when successive output additions fall.
- Negative marginal product means another worker actually reduces total output.

WATCH OUT

Do not wait for marginal product to become negative. It is diminishing as soon as the next worker adds less output than the previous worker.

WORKED EXAMPLE: DIMINISHING MARGINAL PRODUCT

Successive workers and output

	Total product	Marginal product
3 workers	34	Not given
4 workers	46	12
5 workers	56	10

Total product rises from 34 units with three workers to 46 with four, so the fourth worker's marginal product is 12. The fifth worker raises output from 46 to 56, giving marginal product of 10. Output still increases, but the smaller addition shows diminishing marginal product.

CHECK YOURSELF

A sixth worker raises output from 56 to 64. Calculate that worker's marginal product. Is marginal product diminishing, and is total product still rising?